

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 10

Dynamics and Approximation: The Structure of Change

Parallel Reading

Self-Study Curriculum

Parallel Reading: Steven Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed., 2015), Chapters 1-3 (approximately 90 pages, flows on the line and bifurcations). This is the best book ever written for someone with mechanistic intuition who wants to *see* how systems change; it is built on pictures, not proofs. **Core:** Chapters 1-3. **Optional** (if you have the appetite): Chapters 5-6, where the phase plane and the linear-systems picture behind this lecture's Jacobian discussion live, and Chapter 9, where chaos is made visual on the Lorenz equations.

Also read: PCM, section IV.14 ("Dynamics"): a short, map-level overview to place the field relative to the others you have studied.